



Bank Peer Analysis PoC

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Target Stakeholders: Executives, Regulatory Risk.

An interactive data exploration tool built with Streamlit and PyGWalker for analyzing banking institutions' performance and capital metrics using FFIEC data. This PoC was built for financial data analysis and to display rapid prototyping.



Overview

This application serves as a Proof of Concept for financial analysts to filter, visualize, and export bank peer data. Instead of static charts, it integrates an interactive drag-and-drop visualization engine.

Key Features

- **Interactive Explorer:** Uses PyGWalker to provide a Tableau-like interface for ad-hoc data visualization.
- **Dynamic Filtering:** Sidebar controls to filter institutions by Total Assets (in billions).
- **Excel Export:** Clean data extraction of filtered results using `xlsxwriter`.
- **Smart Caching:** Optimized data loading for better performance.



Tech Stack

- **Frontend:** Streamlit
- **Visualization:** PyGWalkerData
- **Handling:** Pandas & XlsxWriter



Project Structure

```
text|— app.py          # Main Streamlit application
    |— data/
    |   |— filtered_bank_data.csv # Input dataset (FFIEC data)
```

```
|— requirements.txt      # Project dependencies
|— README.md
```

Installation & Setup

1. Clone the repository:

```
git clone https://github.com
cd bank-peer-analysis
```

2. Install dependencies:

```
pip install streamlit pandas pygwalker xlswriter
```

3. Prepare your data:

Ensure your bank data is located at `data/filtered_bank_data.csv`. The script expects a column named Total Assets (\$000).

4. Run the app:

```
streamlit run app.py
```

How to Use

- 1. Adjust Thresholds:** Use the sidebar slider to set the minimum asset size (e.g., banks with > \$50B in assets).
- 2. Explore Data:** Use the PyGWalker interface to drag dimensions (Bank Name, State) and measures (Total Assets, ROA) into the shelf to create instant charts.
- 3. Export:** Click the Download Peer List button to save your filtered dataset as an Excel file.



Screenshots

Analysis Settings

Minimum Total Assets (\$ Billions)

10

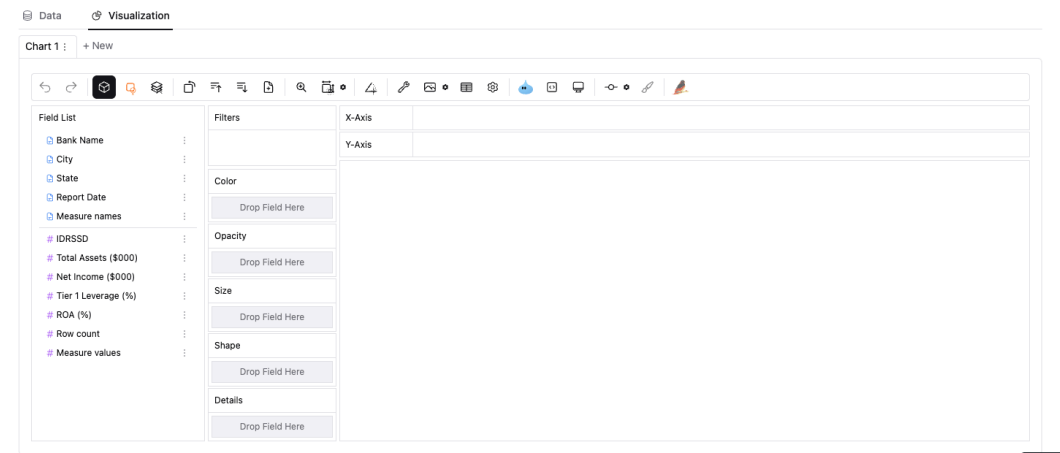
Export Results

Download Peer List (Excel)

Bank Peer Analysis PoC

Performance & Capital Analysis: Top Institutions

Displaying 307 institutions meeting the \$10B threshold.



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Analysis Settings

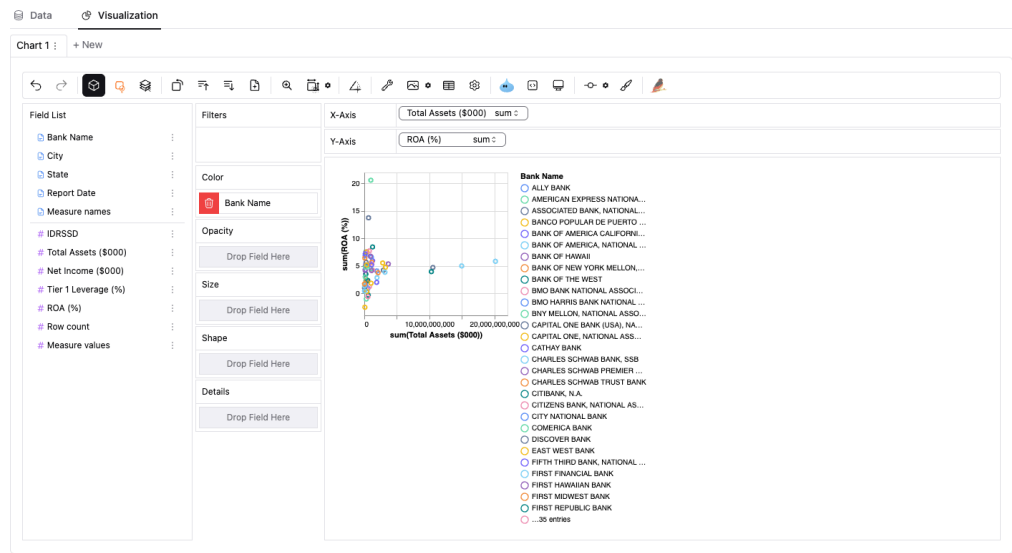
Minimum Total Assets (\$ Billions)

10

Export Results

Download Peer List (Excel)

Displaying 307 institutions meeting the \$10B threshold.



Analysis Settings

Minimum Total Assets (\$ Billions)

490

Export Results

Download Peer List (Excel)

Bank Peer Analysis PoC

Performance & Capital Analysis: Top Institutions

Displaying 45 institutions meeting the \$490B threshold.

DataVisualization

Chart 1 : + New

Field List

Bank Name

City

State

Report Date

Measure names

IDRSSD

Total Assets (\$000)

Net Income (\$000)

Tier 1 Leverage (%)

ROA (%)

Row count

Measure values

Filters

Bank Name

Drop Field Here

Opacity

Drop Field Here

Size

Drop Field Here

Shape

Drop Field Here

Details

Drop Field Here

X-Axis

Total Assets (\$000) sum

Y-Axis

ROA (%) sum

sum(ROA (%))

sum(Total Assets (\$000))

Bank Name

BANK OF AMERICA, NATIONAL ...

CITIBANK, N.A.

GOLDMAN SACHS BANK USA

JPMORGAN CHASE BANK, NAT...

PNC BANK, NATIONAL ASSOCI...

TRUIST BANK

U.S. BANK NATIONAL ASSOCIA...

WELLS FARGO BANK, NATION...

Analysis Settings

Minimum Total Assets (\$ Billions)

480

Export Results

Download Peer List (Excel)

Displaying 47 institutions meeting the \$480B threshold.

DataVisualization

Chart 1 : + New

Field List

Bank Name

City

State

Report Date

Measure names

IDRSSD

Total Assets (\$000)

Net Income (\$000)

Tier 1 Leverage (%)

ROA (%)

Row count

Measure values

Filters

Drop Field Here

Opacity

Drop Field Here

Size

Drop Field Here

Shape

Drop Field Here

Details

Drop Field Here

X-Axis

Bank Name

Y-Axis

Total Assets (\$000) sum

sum(Total Assets (\$000))

Bank Name

BANK OF AMERICA, NATIONAL ASS...

CAPITAL ONE NATIONAL ASSOCIA...

CITIBANK, N.A.

GOLDMAN SACHS BANK USA

JPMORGAN CHASE BANK, NATI...

PNC BANK, NATIONAL ASSOCIATION

TRUIST BANK

U.S. BANK NATIONAL ASSOCIATION

WELLS FARGO BANK, NATIONAL A...

https://github.com/gram99/bank-poc-repo/blob/main/README.md

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